

MGS 616

Assignment 1

Analysis Report



Submitted by -

1. Tanweer Ashif (tanweera@buffalo.edu)
2. Anuja Patil (anujabip@buffalo.edu)
3. Nimisha Nooti (nimishan@buffalo.edu)
4. Saumya Kumari (skumari3@buffalo.edu)
5. Satyam (satyam@buffalo.edu)

	id	host_days	host_response_time	host_response_rate	host_acceptance_rate	host_is_superhost	host_listings_count	host_identity_verified	neighborhood
0	3075044	1669	within an hour	100%	100%	0	1	1	
1	6976	2792	within a few hours	100%	88%	1	1	1	
2	1436513	1351	within a few hours	100%	50%	0	1		0
3	7651065	966	within an hour	100%	100%	1	1	1	
4	12386020	284	within a few hours	100%	95%	1	2		1

5 rows x 33 columns

Overview of the Dataset

`airbnbdf.info()`

By using `airbnbdf.info()` we get a detailed look at the dataset, showing the data types of each column, the count of non-null entries, and basic summary statistics for the numerical columns. It helps in understanding the structure and key characteristics of the dataset.

Data columns (total 33 columns):			
#	Column	Non-Null Count	Dtype
0	id	3583 non-null	int64
1	host_days	3583 non-null	int64
2	host_response_time	3114 non-null	object
3	host_response_rate	3114 non-null	object
4	host_acceptance_rate	3114 non-null	object
5	host_is_superhost	3583 non-null	int64
6	host_listings_count	3583 non-null	int64
7	host_identity_verified	3583 non-null	int64
8	neighbourhood_cleansed	3583 non-null	object
9	city	3581 non-null	object
10	property_type	3580 non-null	object
11	room_type	3583 non-null	object
12	accommodates	3583 non-null	int64
13	bathrooms	3569 non-null	float64
14	bedrooms	3573 non-null	float64
15	beds	3574 non-null	float64
16	bed_type	3583 non-null	object
17	price	3583 non-null	int64
18	security_deposit	3583 non-null	int64
19	cleaning_fee	3583 non-null	int64
20	guests_included	3583 non-null	int64
21	extra_people	3583 non-null	int64
22	minimum_nights	3583 non-null	int64
23	review_scores_rating	2772 non-null	float64
24	review_scores_accuracy	2762 non-null	float64
25	review_scores_cleanliness	2767 non-null	float64
26	review_scores_checkin	2765 non-null	float64
27	review_scores_communication	2767 non-null	float64
28	review_scores_location	2763 non-null	float64
29	review_scores_value	2764 non-null	float64
30	instant_bookable	3583 non-null	int64
31	cancellation_policy	3583 non-null	object
32	reviews_per_month	2829 non-null	float64

dtypes: float64(11), int64(13), object(9)

`airbnbdf.shape()`

The `airbnbdf.shape` function shows that the dataset has 3,583 rows and 33 columns. This gives a clear indication of the dataset's size, with 3,583 observations and 33 features available for analysis.

(3583, 33)

`airbnbdf.dtypes`

The `airbnbdf.dtypes` function provides the data types of each column in the dataset, helping to understand whether the variables are stored as integers, floats, objects (strings), or other types. This helps us to handle each variable during analysis.

id	int64
host_days	int64
host_response_time	object
host_response_rate	object
host_acceptance_rate	object
host_is_superhost	int64
host_listings_count	int64
host_identity_verified	int64
neighbourhood_cleansed	object
city	object
property_type	object
room_type	object
accommodates	int64
bathrooms	float64
bedrooms	float64
beds	float64
bed_type	object
price	int64
security_deposit	int64
cleaning_fee	int64
guests_included	int64
extra_people	int64
minimum_nights	int64
review_scores_rating	float64
review_scores_accuracy	float64
review_scores_cleanliness	float64
review_scores_checkin	float64
review_scores_communication	float64
review_scores_location	float64
review_scores_value	float64
instant_bookable	int64
cancellation_policy	object
reviews_per_month	float64
dtype:	object

Check for Missing Values

In this step missing values were identified which is crucial part of data cleaning. This step calculates the number of missing values in each column, helping to decide how to handle them.

Missing values in the dataset:

id	0
host_days	0
host_response_time	469
host_response_rate	469
host_acceptance_rate	469
host_is_superhost	0
host_listings_count	0
host_identity_verified	0
neighbourhood_cleansed	0
city	2
property_type	3
room_type	0
accommodates	0
bathrooms	14
bedrooms	10
beds	9
bed_type	0
price	0
security_deposit	0
cleaning_fee	0
guests_included	0
extra_people	0
minimum_nights	0
review_scores_rating	811
review_scores_accuracy	821
review_scores_cleanliness	816
review_scores_checkin	818
review_scores_communication	816
review_scores_location	820
review_scores_value	819
instant_bookable	0
cancellation_policy	0
reviews_per_month	754
dtype:	int64

Missing values in the columns relevant to our analysis

```
Missing values in each column:
price                0
beds                 9
bedrooms             10
review_scores_rating 811
dtype: int64
```

Handling Missing Values:

- The **beds** and **bedrooms** columns had 9 and 10 missing values, which were handled by dropping the rows with missing data.
- For the **review_scores_rating** column, with 811 missing values, dropping rows could have compromised the dataset's integrity. Therefore, the missing values were replaced with the **median** to maintain consistency and preserve data quality.

Check for Outliers

Checked for outliers, this step involved generating summary statistics for numerical columns to identify potential outliers. Identified outliers using the IQR method.

Code snippet

Function to Identify Outliers Using the IQR Method

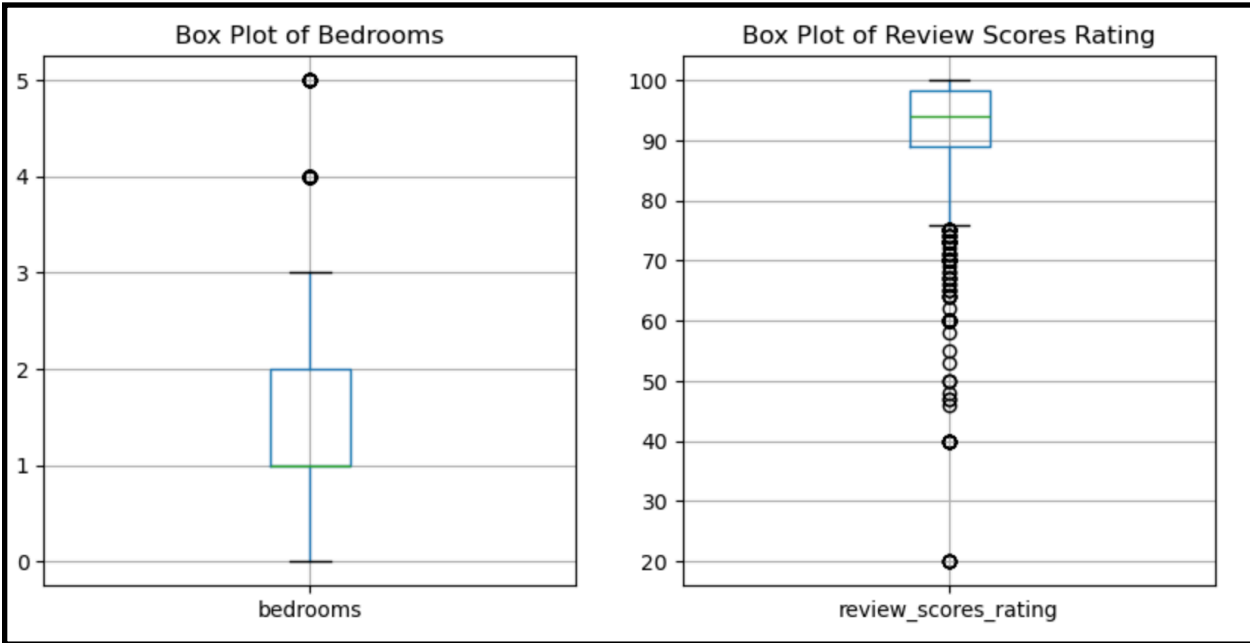
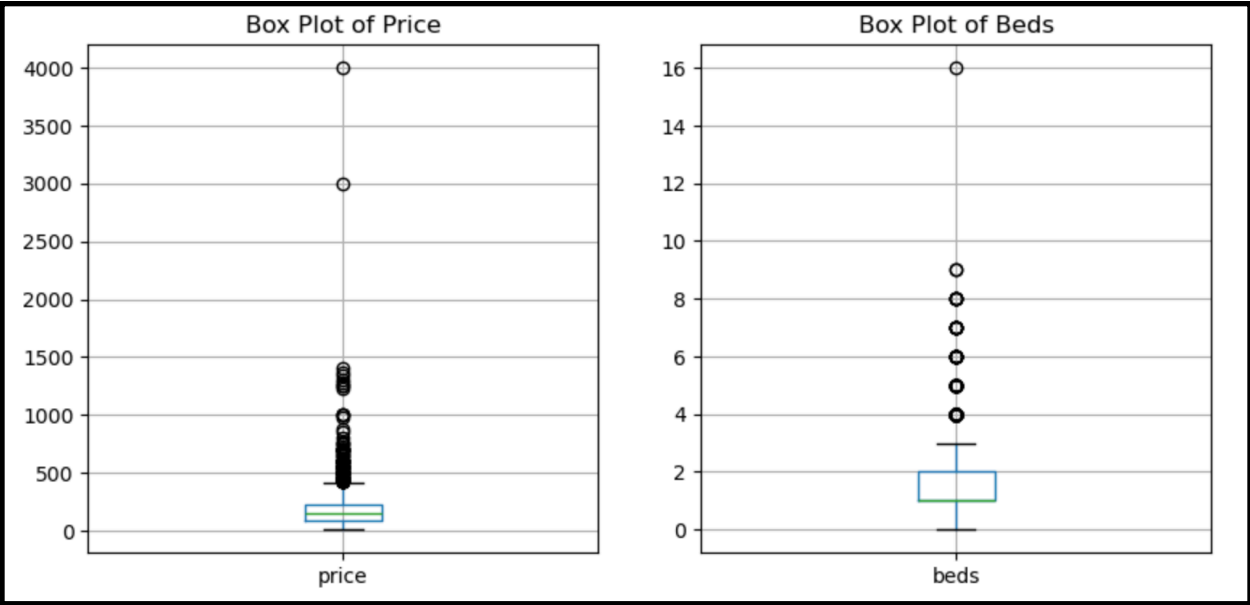
```
def custom_finder(airbnb_df):
    q1=airbnb_df.quantile(0.25)
    q3=airbnb_df.quantile(0.75)
    IQR=q3-q1
    out_l = airbnb_df[((airbnb_df<=(q1-1.5*IQR)) | (airbnb_df>=(q3+1.5*IQR)))]
    return out_l

outl = custom_finder(airbnb_df['price'])
print("Number of outliers in the 'price' column: "+ str(len(outl)))
Number of outliers in the 'price' column: 124

outl = custom_finder(airbnb_df['beds'])
print("Number of outliers in the 'beds' column: "+ str(len(outl)))
Number of outliers in the 'beds' column: 179

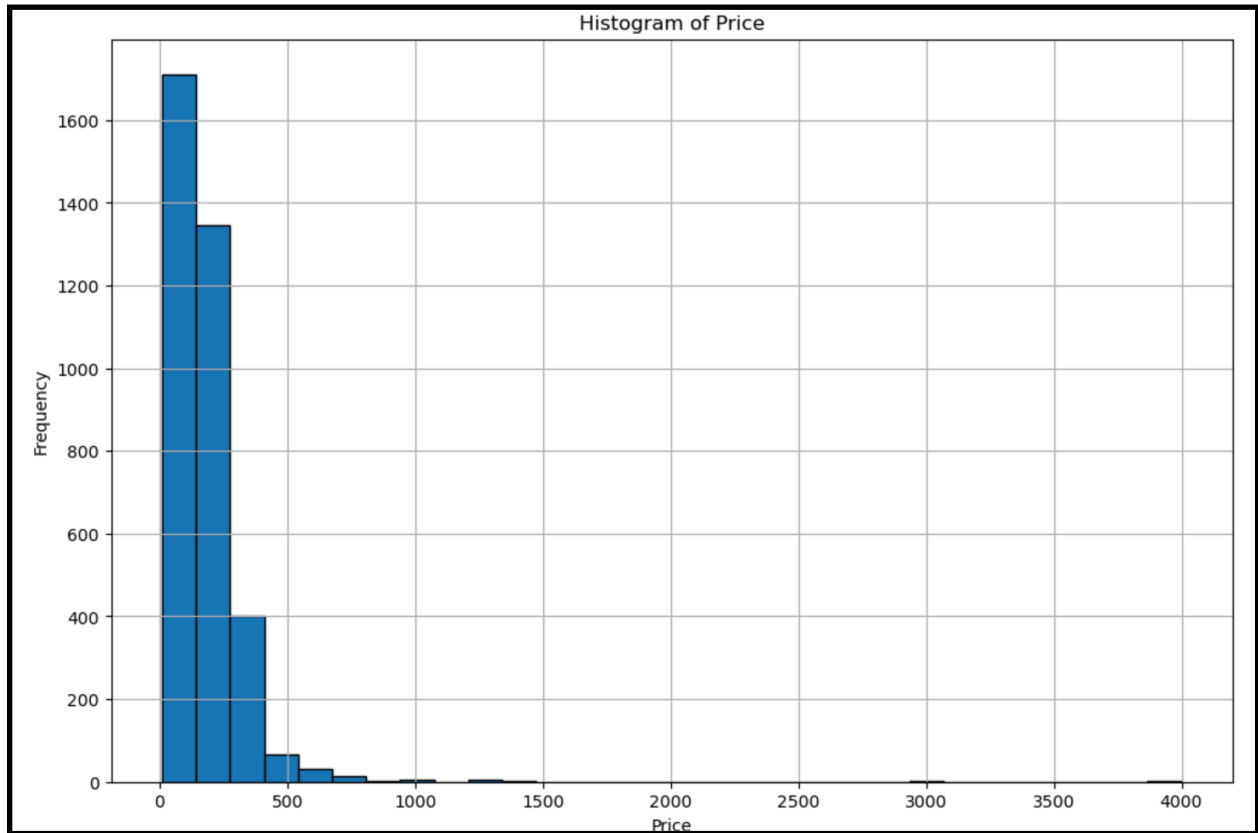
outl = custom_finder(airbnb_df['bedrooms'])
print("Number of outliers in the 'bedrooms' column: "+ str(len(outl)))
Number of outliers in the 'bedrooms' column: 61

outl = custom_finder(airbnb_df['review_scores_rating'])
print("Number of outliers in the 'review_scores_rating' column: "+ str(len(outl)))
Number of outliers in the 'review_scores_rating' column: 137
```



1. Create a histogram for the PRICE variable:

a) Provide the histogram of the variable.

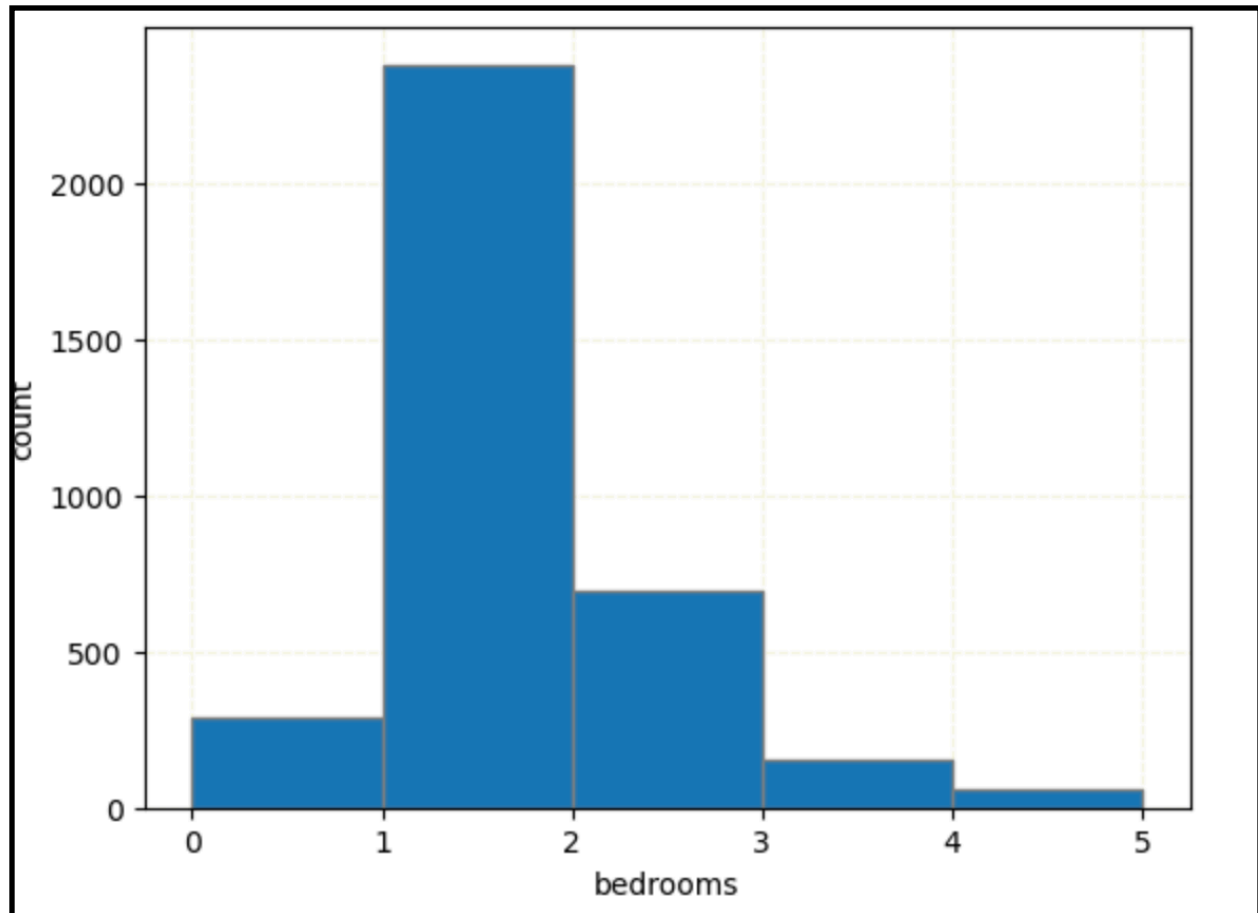


b) Discuss whether transformation is needed for the variable, and why.

- By looking at the histogram, it shows a **right-skewed distribution** for the price variable. Most values are concentrated at the lower end, between 0 and 500, while a few extreme values go up to 4000.
- Yes, it's recommended to transform the **price** variable as it is our target variable since the graph is highly skewed towards right, however, transformation for the data type is not needed for the **price** variable because it's already a numeric data type and has no missing values. Furthermore, there are 124 outliers in the **price** variable that should be handled to stop them from affecting the analysis too much.

2. Create a histogram for the BEDROOMS variable:

a) Provide the histogram of the variable.

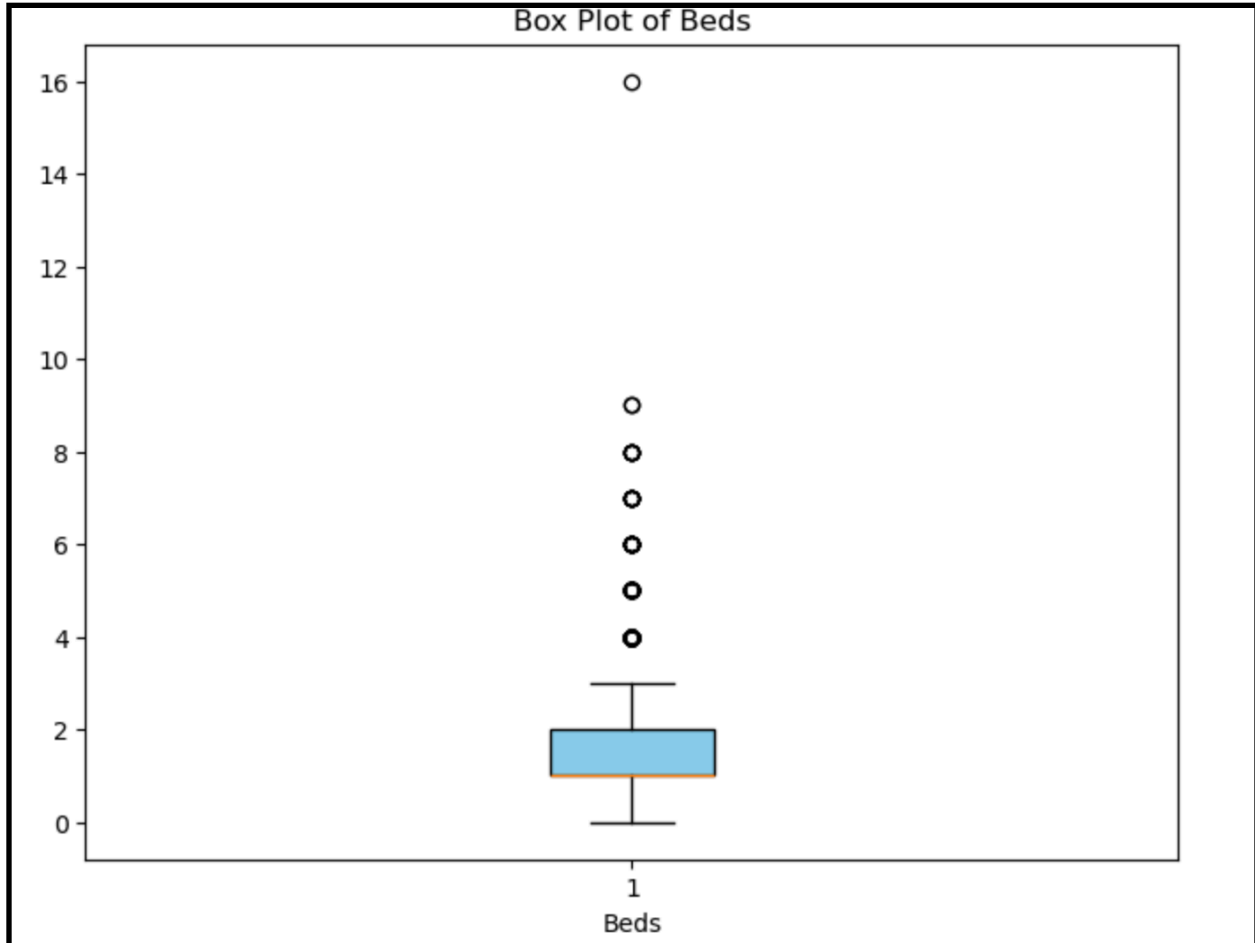


b) Discuss whether transformation is needed for the variable, and why.

- Missing values in the **bedrooms** column were addressed by removing the rows, as only 9 values were missing.
- Additionally, no transformation was necessary because the **bedrooms** variable represents a simple count and bedrooms variable can be of float data type. Also, its distribution is appropriate for the type of analysis being conducted.
- The histogram tells us that the **bedrooms** variable appears to be right-skewed, meaning the majority of properties have 1 or 2 bedrooms, with fewer properties having 3 or more.

3. Create a boxplot for the BEDS variable:

a) Provide the boxplot of the variable.



b) Report the min whisker, max whisker, and mean of the variable

Code Snippet:

```
mean_beds = airbnbdf.beds.mean()

Q1 = airbnbdf.beds.quantile(0.25)
Q3 = airbnbdf.beds.quantile(0.75)

IQR = Q3 - Q1

min_whisker = max(airbnbdf.beds.min(), Q1 - 1.5 * IQR)
max_whisker = min(airbnbdf.beds.max(), Q3 + 1.5 * IQR)

print(f"Minimum Whisker: {min_whisker}")
print(f"Maximum Whisker: {max_whisker}")
print(f"Mean of Beds: {mean_beds}")
```

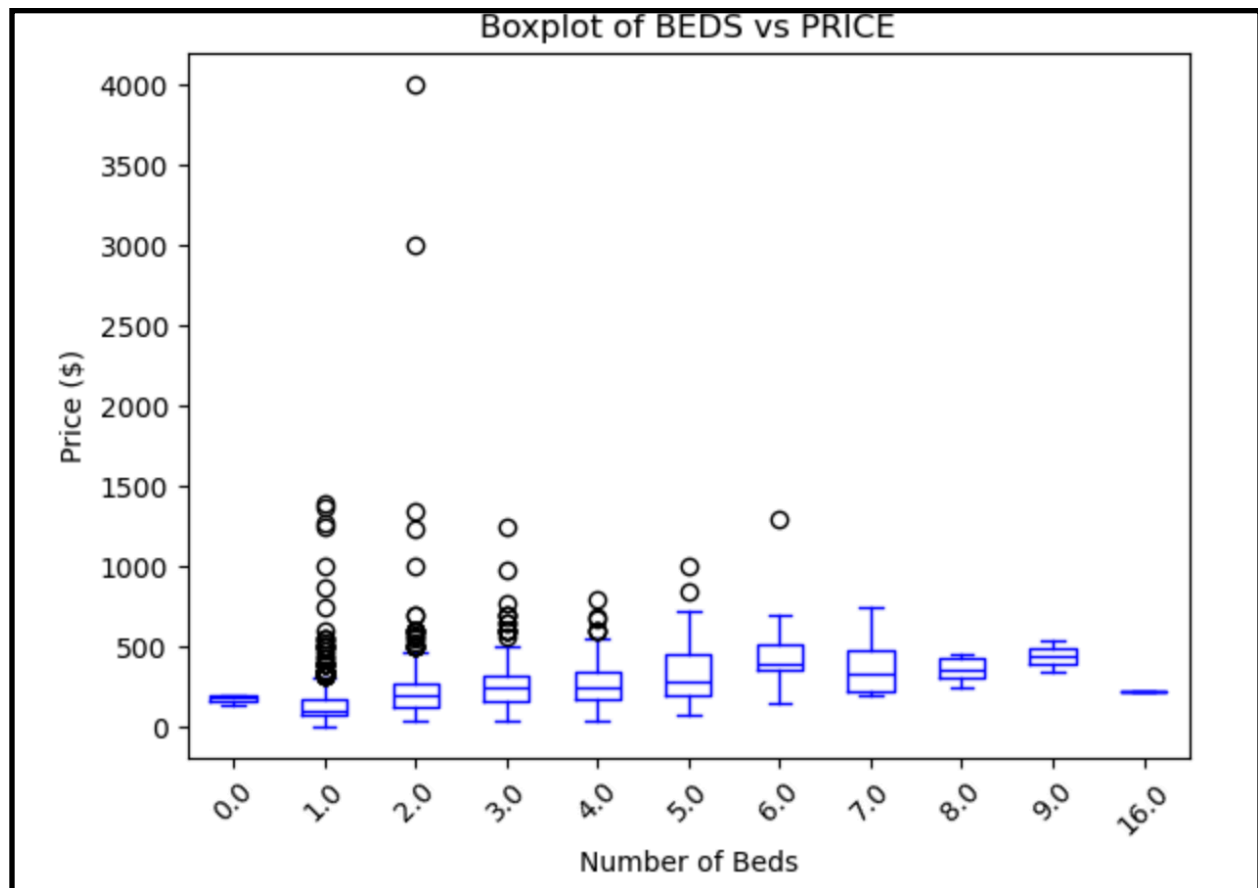
- Minimum Whisker: **0.0**
- Maximum Whisker: **3.5**
- Mean of Beds: **1.61**

c) Discuss whether there are outliers and provide the value of the farthest outlier, if any

- Yes, there are clear **outliers** in the **beds** variable. In the box plot, the circles above the whiskers represent data points that are considered outliers. These are values that fall outside the normal range (above 1.5 times the interquartile range), meaning, the mean number of beds is **1.61**, indicating that most properties have fewer than 2 beds, making the outliers particularly significant in this dataset.
- The farthest outlier in the plot appears to be at **16 beds**, which is significantly higher than most other data points in this variable.

4. Create a boxplot for BEDS (as X) and PRICE (as Y):

a) Provide the boxplot.



b) Discuss two insights you gathered from the boxplot (your insights might concern a relationship, anomaly, skewness, outliers, etc.)

Outliers in Higher Price Ranges:

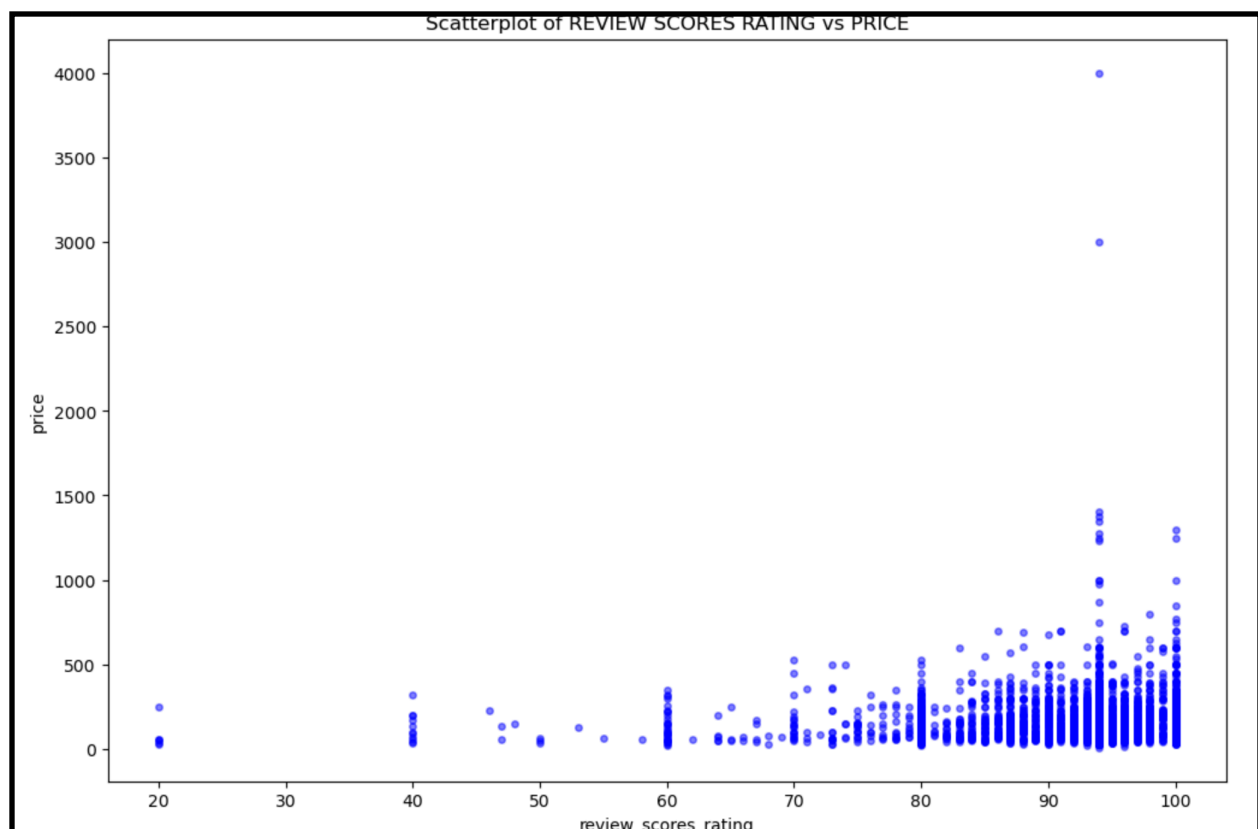
- The boxplot shows several **outliers** for properties with 1 to 4 beds, where some prices are much higher, even reaching up to \$4000.
- Properties with 1 to 4 beds seem to have more price outliers compared to those with more beds. This means that smaller properties can sometimes be priced much higher, likely due to factors like location, amenities, or unique features.

Price Distribution Across Different Numbers of Beds:

- The **median price** generally goes up as the number of beds increases, but the price range also becomes wider.
- For example, properties with more than 4 beds show a bigger spread in prices. However, there's no clear pattern, as some smaller properties with 1 or 2 beds are priced similarly to larger ones. This suggests that other factors like location or luxury features might affect the price, not just the number of beds.

5. Create a scatterplot between PRICE (as Y) and REVIEW_SCORES_RATING (as X):

a) Provide the scatterplot



b) Comment on the observed relationship (if there isn't a discernible relationship, explain why you can't identify a relationship from the chart.)

Based on the scatterplot between **Review Scores Rating** and **Price**, here are a few insights.

No Clear Relationship:

- There doesn't appear to be a strong or clear relationship between **review scores rating** and **price**. The points are widely scattered across the plot, with no clear trend or pattern(e.g no linear trend) that suggests a direct relationship between these two variables.

Price Outliers:

- There are a few listings with very high prices, over \$3000. These high prices don't seem to match any pattern with the review scores, which makes it harder to see any clear relationship between price and review scores.